Failure Path Register

Assignment 1: Student Project Approval & Allocation System (BPMN 2.0). Companion to **student_project_approval.bpmn**. Name / Reg. No.: _____

1. Process scope and happy path

Starts when a student team decides to submit a proposal. **Ends when** a guide is allocated and confirmed and the team is notified, or the proposal is closed as rejected, lapsed, withdrawn or invalid. One pool, five lanes: Student / Team, Project Coordinator, Review Committee (incl. HoD), Faculty Guide, Project Management System (PMS).

#	Step	Lane	BPMN element	Why this type
1	Team decides to submit a proposal	Student	Start event (none)	Process is triggered by the team itself, not by a message or timer.
2	Prepare & submit project proposal	Student	User task	A person fills in the form; the task needs human input.
3	Validate form & team rules	PMS	Service task	Automated check: completeness, roll numbers, team size, existing membership.
4	Proposal valid?	PMS	Exclusive gateway	Exactly one outcome applies: valid, form errors, or team rule violated.
5	Screen proposal (scope, syllabus fit)	Coordinator	User task	Needs the coordinator's judgement.
6	Run similarity / duplicate-topic check	PMS	Service task	Automated comparison against earlier projects.
7	Similarity above threshold?	PMS	Exclusive gateway	Either the topic is original or it goes to the coordinator, never both.
8	Evaluate proposal	Committee	User task	Committee members decide. Carries two non-interrupting timers for reminder and HoD escalation.
9	Committee decision	Committee	Exclusive gateway	One decision only: approved, revise or rejected.
10	Guide allocation & confirmation (sub-process)	PMS / Guide	Expanded sub-process	Groups the allocation attempts so one boundary event can cover withdrawal and no-guide errors for the whole scope.
10a	Check guide availability, workload & domain match	PMS	Business rule task	Allocation is a rule decision (domain match, load cap) that should be changeable without redrawing the process.
10b	Eligible guide found?	PMS	Exclusive gateway	A guide exists or does not.
10c	Allocate guide; Send allocation request to guide	PMS	Service task; Send task	Reserving the slot is automated; the request is a one-way message to the guide.
10d	Review request & accept / decline	Guide	User task	Guide decides. 5-day timer boundary covers silence.
10e	Guide response	Guide	Exclusive gateway	Accepted or declined.
11	Record allocation in PMS; Notify team & guide (sub-process)	PMS	Service task; Send task	System write, then message to team and guide. Wrapped so that either failure is handled by one retry loop.
12	Project allocated & student notified	PMS	End event (none)	Successful completion.

2. Gateway and event choices

Construct	Where used	Reason
Exclusive gateway (split)	Validation, similarity, coordinator decision, committee decision, final decision, guide eligibility, guide response, all "attempts left?" counters	Each is a single either/or decision. No two outcomes can be true together.
Exclusive gateway (merge)	Entry to submission, entry to committee evaluation, entry to guide allocation, rejection handling, join before record & notify	Several loop-back and normal paths converge on one step. Only one token arrives at a time, so no synchronisation is needed.
Parallel / inclusive gateway	Not used	The process is a sequence of approvals. Nothing in the scenario runs in parallel, and no outcome needs several branches at once. Using them would be forced.

Construct	Where used	Reason
Timer boundary event (interrupting)	Submission deadline; 7-day revision window; 5-day guide response	The wait is cut off and the flow moves to a recovery or closing path.
Timer boundary event (non-interrupting)	Committee review: 5-day reminder, 10-day HoD escalation	The reviewers keep working while the reminder or escalation happens alongside.
Error boundary / error end event	No eligible guide; record or notify failure	A technical or resource failure inside a scope is thrown as an error and caught by the parent scope.
Message boundary event	Withdrawal by team or guide on the allocation sub-process	Withdrawal arrives from outside at any moment during allocation, so it must interrupt the scope.
Compensation (boundary, throw, handler)	Allocate guide, Release guide slot, Undo allocation	A guide slot that was reserved must be given back when the guide declines or stays silent.
Intermediate message catch event	Waiting list: "Guide slot released"	The team waits for an outside signal, not for a fixed time.
Script task	Count attempt, Count revision cycle, Exclude guide & count attempt, Count retry	Pure counter updates that make every loop bounded.

3. Failure Path Register

Every row maps to elements in the diagram. "Where in the model" names the task, event or gateway as labelled in the .bpmn file.

ID	Failure point	Cause / trigger	Where in the model	Solution designed in the diagram	BPMN constructs	End state / outcome
F1	Incomplete or invalid form	Missing fields, no abstract, invalid roll numbers.	Validate form & team rules, then "Proposal valid?" (Form errors branch).	Count attempt, then "Attempts left?". Under 2 attempts: proposal is returned with an error list and the team corrects it. At 2 attempts: coordinator resolves it manually.	Exclusive gateway, script task counter, send task, loop to submission, escalation to a user task.	Loop back to submission, or Closed: invalid submission.
F2	Team rule violation	Team size over the limit, or a student already belongs to another team.	"Proposal valid?" (Team rule violated branch).	Team is told to reform. No retry loop, because a rule breach is not a typing error.	Exclusive gateway, send task, end event.	Closed: invalid team.
F3	Submission deadline missed	Team does not submit by the deadline.	Boundary timer on "Prepare & submit project proposal".	Coordinator decides on a late-submission request. If approved, the team returns to submission; if not, the case closes.	Interrupting timer boundary, user task, exclusive gateway.	Loop to submission, or Closed: deadline missed.
F4	Duplicate or too-similar topic	Similarity score above the threshold against earlier projects.	"Similarity above threshold?" then Review similarity match and "Coordinator decision".	Coordinator chooses: accept (go on to the committee), request a modified topic (send task, loop to submission), or reject as duplicate.	Service task, exclusive gateways, send task, loop, merge into rejection handling.	Continue, loop, or Closed: rejected.
F5	Committee asks for revision	Scope too large, objectives vague, weak novelty.	"Committee decision" (Revise branch), "Revision cycles < 2?".	Count revision cycle. Under the limit: revision request goes to the team, who revise and resubmit through validation again. Limit reached: see F6.	Exclusive gateways, script task counter, send task, user task, bounded loop.	Revised proposal re-enters submission.
F6	Revision limit reached	Two revision cycles already used.	"Revision cycles < 2?" (No branch), Issue final decision.	Committee must approve or reject. No further revision is possible.	Exclusive gateways, user task.	Approved into allocation, or into rejection handling.
F7	Revision window expires	Team does not return the revised proposal in 7 days.	Boundary timer on "Revise proposal per committee comments".	The case is closed so it does not stay open forever.	Interrupting timer boundary, end event.	Closed: lapsed (no revision).
F8	Committee rejects proposal	Not feasible, out of syllabus, lacks resources. Also reached from duplicate rejection and final rejection.	"Committee decision" (Rejected branch); merge before "Send rejection reasons to team".	Reasons are sent. First rejection only: the team may submit one new topic (loop to submission). Otherwise the case closes.	Exclusive merge gateway, send task, exclusive gateway "New topic allowed?", loop.	New submission, or Closed: rejected.
F9	Committee review delayed	No decision within the review SLA.	Two timers on "Evaluate proposal".	After 5 days a reminder is sent. After 10 days the HoD steps in to expedite. Both run while the review continues.	Non-interrupting timer boundary events, send task, user task.	Review continues; side branches end as Reminder sent and Escalation handled.
F10	No eligible guide	No domain match, or every matching guide is at maximum workload.	"Eligible guide found?" (No) and error boundary on Guide allocation & confirmation.	Error thrown to the sub-process boundary. Coordinator allocates manually (co-guide, external guide) or places the team on a waiting list and waits for a released slot.	Business rule task, error end event, error boundary event, user task, exclusive gateway, message catch event, loop.	Guide assigned, or waits and re-enters allocation.
F11	Guide declines or does not respond	Workload, conflict of interest, or silence for 5 days.	Boundary timer on "Review request & accept / decline"; "Guide response" (Declined).	Guide is excluded and the attempt counted. The reserved slot is released by compensation, then "Attempts < 3?" loops back to the rule check. After 3 attempts an error is raised and F10 manual handling takes over.	Timer boundary, compensation boundary and handler, compensation throw event, script task, exclusive gateway, error end event.	Next guide tried, or manual allocation.

ID	Failure point	Cause / trigger	Where in the model	Solution designed in the diagram	BPMN constructs	End state / outcome
F12	Withdrawal after approval	Team dissolves, or the guide withdraws or goes on leave.	Message boundary on Guide allocation & confirmation; "Who withdrew?".	Withdrawal is recorded and the guide slot released. If the team withdrew, the case closes. If the guide withdrew, the team goes back into allocation.	Message boundary event (interrupting), user task, exclusive gateway, loop.	Closed: team withdrew, or re-allocation.
F13	Record or notification failure	PMS write error, or email / notification service down.	Error boundaries on Record allocation in PMS and Notify team & guide, and on the Record allocation & notify sub-process.	Count retry. Retry up to 3 times; after 3 failures the coordinator notifies the team manually and logs the failure.	Error boundary events, error end event, script task counter, exclusive gateway, loop, user task.	Project allocated, or Project allocated (manual notice).

4. Loop limits and end states

Loop	Limit	What happens at the limit
Form correction (F1)	2 attempts	Escalated to the coordinator
Revision cycles (F5)	2 cycles	Committee must approve or reject
New topic after rejection (F8)	1 new topic	Closed: rejected
Guide attempts (F11)	3 guides	Manual allocation by the coordinator
Notification retries (F13)	3 retries	Manual notification by the coordinator

End events in the model. Success: Project allocated & student notified; Project allocated (manual notice). Closed: deadline missed, invalid submission, invalid team, rejected, lapsed (no revision), team withdrew. Non-process ends: Reminder sent, Escalation handled, Guide confirmed, plus error ends that are caught by a boundary event (No eligible guide, Guide attempts exhausted, Record / notify error).